2 Ozone, O_3, plays a crucial role in the Earth's atmosphere by absorbing harmful ultraviolet radiation. It is also widely used for its oxidising and disinfecting properties. For example, ozone can be dissolved in ground water or drinking water for disinfection and water quality enhancement.

Fig. 2.1 shows one possible structure of O_3 (g).

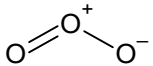


Fig. 2.1

(a) The overall reaction for the decomposition of ozone can be represented as follows.

$$2O_3 \rightarrow 3O_2$$

The rate of decomposition of ozone in ground water, at pH 8, was investigated and the following results were obtained. The reaction is first order with respect to ozone.

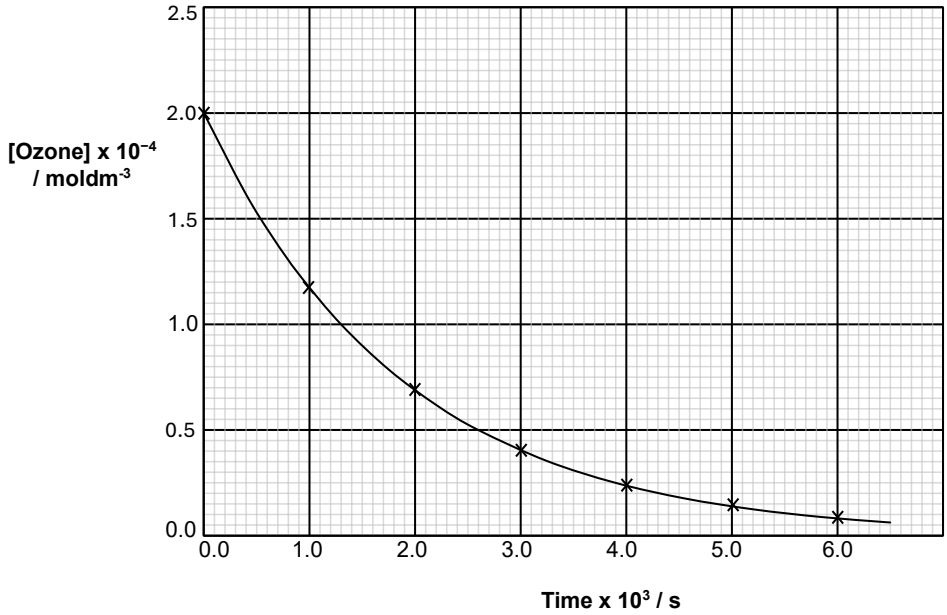


Fig. 2.2

(i) Define the term *order of reaction*.

Commented [KCH1]: Define what is meant by the term "order of reaction"?

.....

..... **[1]**

(ii) Use the graph in Fig. 2.2 to show that the overall order of reaction is first order.

.....

.....

..... **[1]**

(iii) Hence, calculate the value of the rate constant, *k*. Include its units.

[1]

(iv) The presence of OH^- was found to initiate the decomposition of ozone and the following reaction mechanism was suggested.

Step 1: $O_3 + OH^- \rightarrow HO_2^- + O_2$

Step 2: $HO_2^- + O_3 \rightarrow 2O_2 + OH^-$

State the role of OH^- in this mechanism and explain how the presence of OH^- would affect the rate of the reaction.

.....

.....

..... **[1]**

(b) Ozone is a strong oxidising agent, useful for oxidative cleavage of alkenes to form carbonyl compounds.

The reaction of ozone with alkenes can be shown in Fig. 2.3.

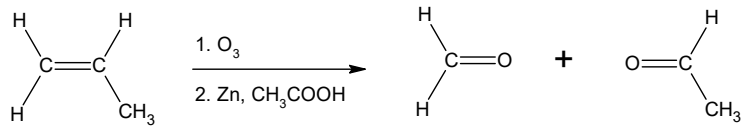


Fig. 2.3

The first step in the mechanism is the initial electrophilic attack by ozone to the carbon-carbon double bond, which then forms the molozonide intermediate. In the second step, the unstable molozonide intermediate undergoes further reaction and breaks apart to form a carbonyl oxide and a carbonyl compound.

The first and second step of the mechanism is shown in Fig. 2.4.

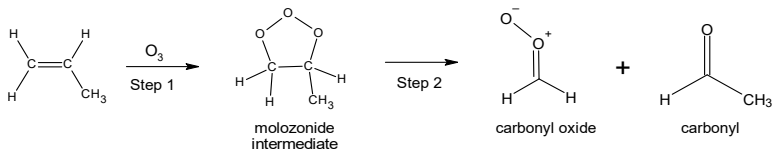


Fig. 2.4

The carbonyl oxide and carbonyl then further react to form the respective carbonyl compounds.

(i) 2-methylbut-2-ene reacts with ozone in a similar reaction to that in Fig. 2.3.

On Fig. 2.5, draw the structure of the molozonide intermediate and suggest the mechanism for the reaction of 2-methylbut-2-ene with ozone in step 1 to form the molozonide intermediate. Include all relevant lone pairs and three curly arrows.

Commented [SXF2]: On Fig 2.5

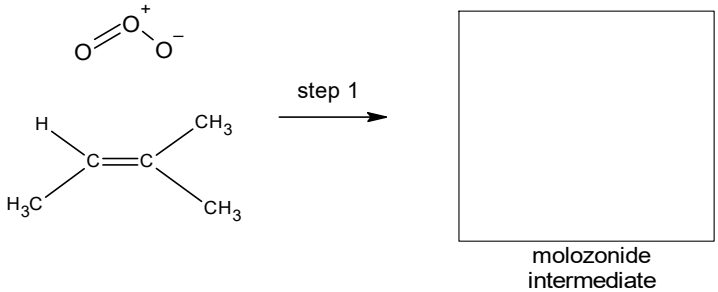
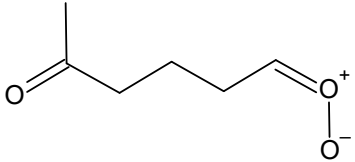


Fig. 2.5 **[2]**

(ii) Compound **F** was formed in step 2 of the mechanism in Fig 2.4 when ozone reacts with another alkene, **G**. Suggest the identity of **G**.



Compound **F**

G

[1]

[Total: 7]